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B.Tech. Degree VI Semester Regular/Supplementary Examination June 2023

ME 19-205-0607 ENERGY CONSERVATION AND ENVIRONMENT PROTECTION (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Understand the overview of World Energy Scenario and energy economics.
 CO2: Study the importance of energy management and energy auditing.
 CO3: Learn about the renewable energy sources.
 CO4: Understand the environmental impacts of energy use.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create
 PO – Programme Outcome

PART A(Answer **ALL** questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) What are the benefits of an energy balance analysis?	3	3	L2	2	7
(b) What are the drawbacks of a Simple Pay Back Period method?	3	3	L2	1	7
(c) What do you mean by flash steam? How it's being used as an energy efficiency improvement tool?	3	3	L2	2	7
(d) How does an oversized pump affect the energy efficiency?	3	3	L2, L3	2	7
(e) Write the concept of Pinch in Heat Exchanger.	3	3	L2	1	7
(f) Name and briefly discuss the different renewable energy sources.	3	3	L1	3	1,7,12
(g) List the various steps to reduce particulates and water pollution.	3	3	L2	4	7,10
(h) Explain how CO ₂ emission causes global warming.	3	3	L2	4	7,10

PART B

(4 × 12 = 48)

II. (a) Analyze the challenges and opportunities in transitioning to a sustainable and renewable energy system in the context of the current world energy scenario.	6	6	L4	1	7
(b) Assess the potential of renewable energy adoption in India and analyze the challenges and strategies for achieving a low carbon energy future in the country.	6	6	L4	1	7

OR

III. A company is considering investing in a new machine that will cost ₹10 lakh. The machine is expected to generate annual cash flows of ₹2 lakh for the next 5 years. The company's discount rate is 10%.	12	12	L5	2	7
(i) What is the NPV of this investment?					
(ii) Find the approximate IRR for this investment.					
(iii) State the acceptance/rejection status for NPV and IRR models.					



(P.T.O.)

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		Marks	BL	CO	PO																				
IV.	(a) An energy audit firm is hired to spot the inefficiencies in the boiler and steam distribution system of their client industry. Describe potential opportunities for saving energy in this context.	8	L3	2	7																				
	(b) What are the different methods to recover waste heat?	4	L1	2	7																				
OR																									
V.	(a) Discuss the potential energy saving opportunities in refrigeration and air conditioning systems.	6	L3	2	7																				
	(b) Explain how past plant data analysis helps energy management.	6	L1	2	7																				
VI.	(a) Define cogeneration. Explain with the help of neat sketch, classification and types of cogeneration.	6	L2	2	7																				
	(b) Explain how heat recovery can be calculated with the help of composite curves. Explain with neat figures.	6	L2	1	7																				
OR																									
VII.	Construct Heat Exchanger Network with the help of Pinch Analysis for the data given below:	12	L4	1	7																				
<table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"><tr><th>Streams</th><th>Specific Heat, Cp</th><th>Source Temperature Ts (°C)</th><th>Target Temperature Tt (°C)</th></tr><tr><td>Hot Stream 1</td><td>2</td><td>150</td><td>60</td></tr><tr><td>Hot Stream 2</td><td>8</td><td>90</td><td>60</td></tr><tr><td>Cold Stream 1</td><td>2.5</td><td>20</td><td>125</td></tr><tr><td>Cold Stream 2</td><td>3</td><td>25</td><td>100</td></tr></table>						Streams	Specific Heat, Cp	Source Temperature Ts (°C)	Target Temperature Tt (°C)	Hot Stream 1	2	150	60	Hot Stream 2	8	90	60	Cold Stream 1	2.5	20	125	Cold Stream 2	3	25	100
Streams	Specific Heat, Cp	Source Temperature Ts (°C)	Target Temperature Tt (°C)																						
Hot Stream 1	2	150	60																						
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Cold Stream 1	2.5	20	125																						
Cold Stream 2	3	25	100																						
VIII.	(a) Describe environmental auditing. Explain its significance.	4	L2	2	7																				
	(b) Explain air pollution, pollutants and their impacts.	5	L2	4	7,10																				
	(c) What are the sources of SO _x , NO _x and CO emissions?	3	L2	4	7,10																				
OR																									
IX.	(a) Give a brief description about different sources of emissions that pollute the environment.	4	L2	4	7,10																				
	(b) Explain the detailed procedure of exhaust emission test in a vehicle.	8	L2	4	7,10																				

Bloom's Taxonomy Levels

L1 = 13%, L2 = 50%, L3 = 11%, L4 = 18%, L5 = 8%.
